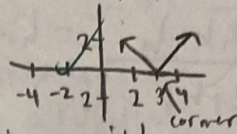


Total
24.5 / 30

1. Analyze the differentiability of $y = |x - 3|$. Include a diagram and use proper terminology.

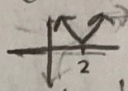
$$y = \begin{cases} x-3, & x \geq 3 \\ -(x-3) = -x+3, & x < 3 \end{cases}$$



3 / 13

This is an absolute value function with a corner (a point where it is not differentiable) at $x=3$. Otherwise it is differentiable for the rest of the function.

2. True or False: The domain of a function $f(x)$ and the domain of its derivative $f'(x)$ are always equal. Provide an example and use terminology discussed in this course in your explanation.

False, we know that a function can be continuous throughout its domain but not always differentiable (v.t., cusp, corner). This means that while a function like  (corner) may have a domain of $\{x \in \mathbb{R}\}$, its derivative's domain is $\{x \in \mathbb{R} \mid x \neq 2\}$.

3. Determine the equation of the normal to the function $f(x) = \sqrt{2x+5}$ at $x=1$. $f(1) = \sqrt{2+5} = \sqrt{7}$

$$f'(x) = \frac{1}{2} (2x+5)^{-\frac{1}{2}} \cdot 2 = \frac{1}{\sqrt{2x+5}} \quad \text{perpendicular tangent}$$

$$f'(1) = \frac{1}{\sqrt{2+5}} = \frac{1}{\sqrt{7}} \quad \text{take neg. rec.}$$

$$y = -\sqrt{7}x + b$$

$$\sqrt{7} = -\sqrt{7}(1) + b$$

$$b = 2\sqrt{7}$$

$$y = -\sqrt{7}x + 2\sqrt{7}$$

13

4. What is the x-intercept of the line tangent to the curve $f(x) = 4x^3 + 2x - 3$ at the point $(1, 3)$?

$$f'(x) = 12x^2 + 2$$

$$y = 10x + b \quad y = 10x - 7$$

$$f'(1) = 12(1)^2 + 2$$

$$3 = 10(1) + b$$

$$= 12 + 2$$

$$b = 3 - 10$$

$$= 14$$

$$b = -7$$

$$0 = 10x - 7$$

$$7 = 10x$$

$$x = \frac{7}{10}$$

check

$$y = 10\left(\frac{7}{10}\right) - 7$$

$$= 7 - 7$$

$$= 0$$

$$\therefore \text{x-int} = \left(\frac{7}{10}, 0\right)$$

5. If $f(x) = \frac{k+x}{1+x}$ and $\lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = 2$, determine the value of k .

$$2 = \lim_{h \rightarrow 0} \left(\frac{k+1+h}{2+h} - \frac{k+1}{2} \right) \frac{1}{h} = \lim_{h \rightarrow 0} \frac{2k+2h+2 - (k+1)(2+h)}{2h(2+h)} = \lim_{h \rightarrow 0} \frac{2k+2h+2 - (2k+2h+k+1)}{2h(2+h)}$$

$$= \lim_{h \rightarrow 0} \frac{2k+2h+2 - 2k-2h-k-1}{2h(2+h)} = \lim_{h \rightarrow 0} \frac{2-k-1}{2(2+h)} = \frac{1-k}{4}$$

$$\frac{1-k}{4} = 2 \implies 1-k = 8 \implies k = -7$$

6. Find the equation(s) of the tangent(s) from $(3, 18)$ to $y = -x^2 + 6x + 5$.

$\frac{dy}{dx} = -2x + 6$ slope = $\frac{y-18}{x-3} = \frac{-x^2+6x+5-18}{x-3}$

$-2x+6 = \frac{-x^2+6x+5-18}{x-3}$

$(x-3)(-2x+6) = -x^2+6x-13$

$-2x^2+6x+6x-18 = -x^2+6x-13$

$-x^2+6x-5 = 0$

$-(x^2-6x+5) = 0$

$(x-1)(x-5) = 0$

$x=1, x=5$

slopes (1 and 5) these are the slopes

$y = x + b$

$18 = 3 + b$

$b = 15$

$y = x + 15$

$y = 5x + b$

$18 = 5(3) + b$

$18 = 15 + b$

$b = 3$

$y = 5x + 3$

7. The manager of an appliance manufacturing firm determines that when A amount of toasters are priced at p dollars apiece, the number sold each month can be modelled by $A(p) = \frac{300}{p}$. The

manager estimates that t months from now, the unit price of the toasters will be $p(t) = 0.04t^2 + 18.5$ dollars. Using Leibniz notation, determine at what rate the monthly demand for toasters, $A(p)$, will be changing two years from now? Round to two decimal places.

In 2 years, $t = 2 \cdot 12 = 24$

$p(24) = 0.04(24)^2 + 18.5$

$= 23.04 + 18.5$

$= 41.54$ dollars

$\frac{dA}{dp} = \frac{0(1) - 300(1)}{p^2} = -\frac{300}{p^2}$

$= -\frac{300}{p^2}$

$A'(41.54) = -\frac{300}{41.54^2} = -0.17$

$\therefore$ In 2 years, the rate of change for the monthly demand of toasters will be -0.17 units?

8. Find the area of the triangle formed by the 3 points where the horizontal tangent lines exist to the curve $y = x^4 - 2x^2 + 4$. A rough sketch might help.

$$\frac{dy}{dx} = 4x^3 - 4x = f'(x)$$

$$0 = 4x^3 - 4x$$

$$0 = 4x(x^2 - 1)$$

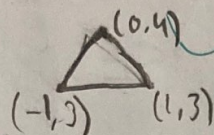
$$0 = 4x(x-1)(x+1)$$

$$x = 0, 1, -1$$

$$f(0) = 4$$

$$f(1) = 1 - 2 + 4 = 3$$

$$f(-1) = 1 - 2 + 4 = 3$$



$$\text{height} = 4 - 3 = 1$$

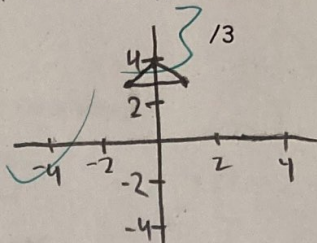
$$\text{base} = 1 - (-1) = 2$$

3 local max/min (slope of 0)

$$(0, 4)$$

$$(1, 3)$$

$$(-1, 3)$$



$$\text{Area of } \Delta = \frac{bh}{2} = \frac{2(1)}{2} = 1 \text{ unit squared}$$

9. Suppose $f(2) = -3$, $g(2) = 4$, $f'(2) = -2$, $g'(2) = 7$. Find $h'(2)$ if $h(x) = \frac{g(x)}{(1+f(x))}$. $\rightarrow \frac{d}{dx}(1+f(x)) = (0+f'(x)) = f'(x)$

$$h'(x) = \frac{g'(x)(1+f(x)) - g(x)(f'(x))}{(1+f(x))^2}$$

$$h'(2) = \frac{g'(2)(1+f(2)) - g(2)(f'(2))}{(1+f(2))^2}$$

$$= \frac{7 \cdot -2 - 4 \cdot -3 \cdot -2}{(-2)^2} = \frac{-14 - 24}{4} = \frac{-38}{4} = \boxed{-\frac{19}{2}}$$

10. Given $x^3 f(x) + (f(x))^3 + f(x^3) = 3$ and $f(1) = 2$, find $f'(1)$.

$$f(x) = a \quad f'(x) = b \quad f'(x^3) = c$$

$$h(x) = 3 \quad \frac{d}{dx}(x^3 f(x)) = 3x^2 f(x) + x^3 f'(x)$$

$$\frac{d}{dx}((f(x))^3) = 3(f(x))^2 (f'(x)) \quad \frac{d}{dx}(f(x^3)) = f'(x^3)$$

$$h'(x) = 3x^2 f(x) + x^3 f'(x) + 3f'(x)(f(x))^2 + f'(x^3)$$

$$0 = 3ax^2 + bx^3 + 3ba^2 + c(x^3)$$

$$bx^3 - 3ba^2 = 3ax^2 - 3ba^2 = -ax^3 - a^3 + 3$$

$$b = \frac{3ax^2 - 3ba^2 - (-ax^3 - a^3 + 3)}{x^3 - 3a^2}$$

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